Harem pants · size 122

harem (wide, dropped crotch) fit, Sweat / French Terry (25 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

Files

- PDF A4 tiles to glue (12 pages + overview): `harem_pants_size122_a4_tiled.pdf`
- Preview A4 tiles: `harem_pants_size122_a4_preview.png`
- SVG master 1:1: `harem_pants_size122_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

Body measurements

Body measurement	Value	Source
Body height	122.0 cm	table
Chest	62.0 cm	table
Waist	56.0 cm	table
Hip	66.0 cm	table
Head	52.0 cm	table
Neck	28.5 cm	table
Inseam	54.0 cm	table
Side length	72.0 cm	table
Arm length	47.0 cm	table
Back length	29.5 cm	table
Shoulder width	10.0 cm	table
Thigh	37.0 cm	table
Ankle	17.8 cm	table

Finished measurements (virtually sewn)

Finished measurement	Value
Hip width	85.8 cm
Thigh width	55.5 cm
Leg width at mid-leg	42.0 cm
Leg opening	20.0 cm
Waistband relaxed	50.4 cm
Waistband stretched (pants top edge)	83.0 cm
Outseam	75.5 cm
Inseam	48.2 cm
Front rise	27.3 cm
Back rise	29.1 cm
Waistband height	3.0 cm
Cuff height	7.0 cm

Finished measurement	Value
Crotch seam total	64.2 cm

Pattern pieces

Piece	Cutting	Fabric	Seam allowance
Front leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Back leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Waistband (elastic)	cut 1 on fold	Main fabric	seam allowance 1.0 cm included
Leg cuff	cut 2	Rib knit	seam allowance 1.0 cm included

Fabric needed

- **Main fabric:** about 0.90 m at 150 cm width (folded selvage to selvage, cut on fold pieces at the fold, incl. 10 % for shrinkage; single layer would need 0.90 m)
- **Rib knit:** about 0.25 m at 70 cm width (single layer, incl. 10 % for shrinkage)
- Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

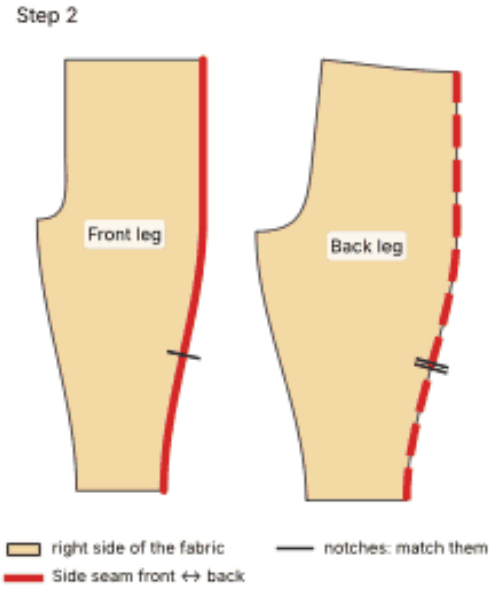
Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

Cut on fold (Waistband (elastic)): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. On the folded fabric, lay the fold edge exactly on the fabric fold, pin it and cut through both layers; opened out you get the whole piece. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

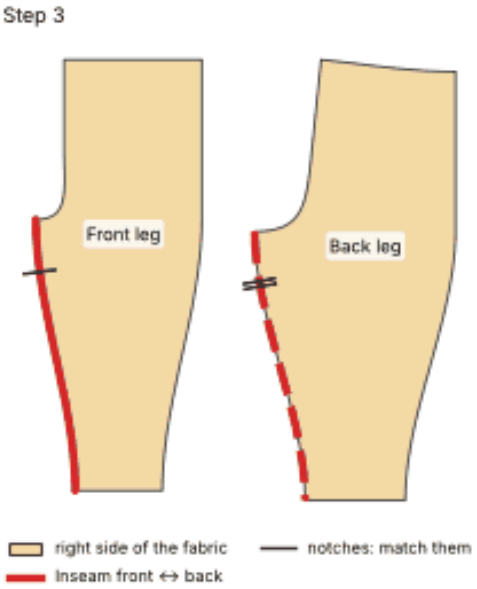
Sewing instructions

Seam allowance is included in the pattern (see table). Use a jersey/stretch needle for jersey and sweat.

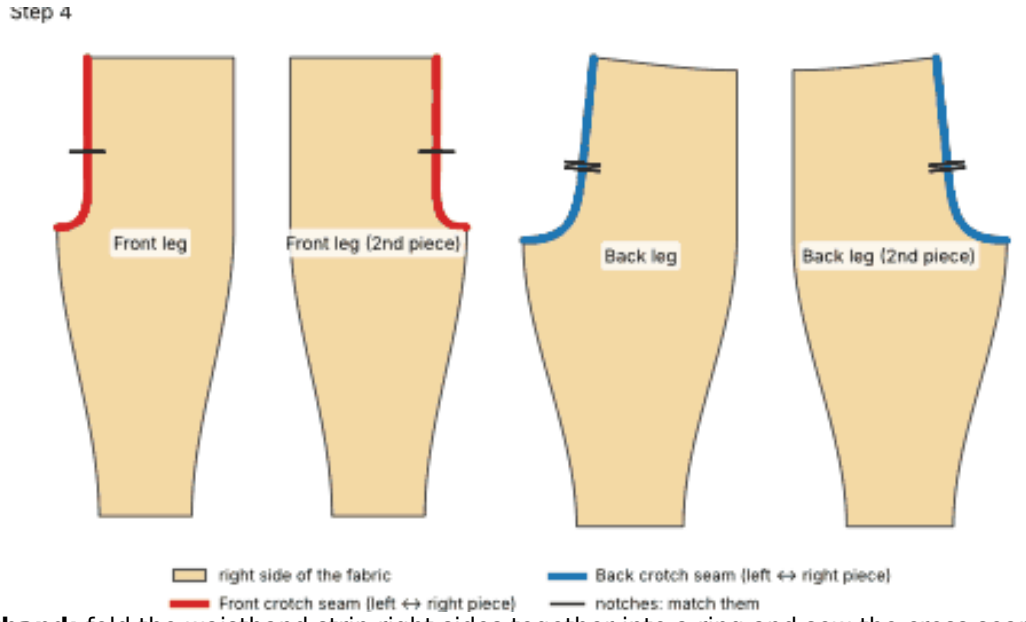
1. **Prepare:** pre-wash the fabric, cut out the pieces, snip the notches 3–4 mm deep.
2. **Side seams:** place front and back legs right sides together (notches: single = front, double = back) and sew the side seams with an overlocker or a narrow zigzag/stretch stitch.



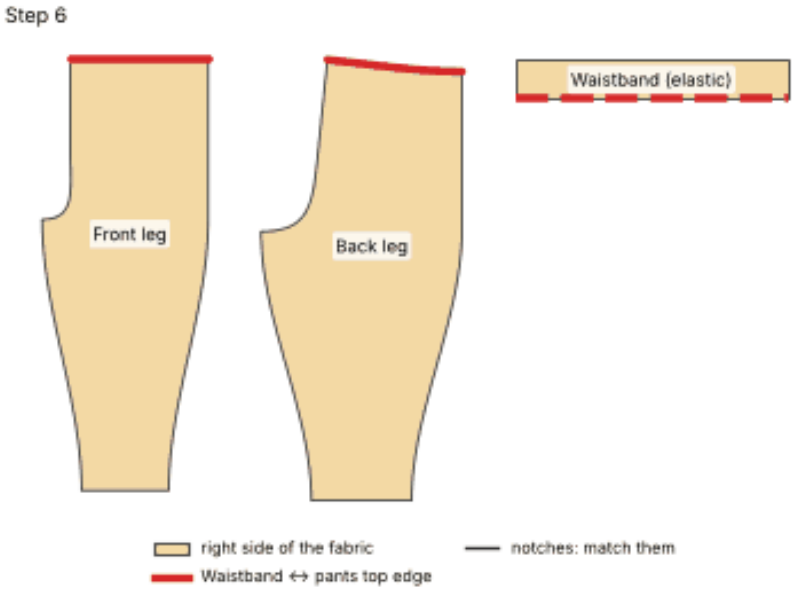
3. **Inseams:** sew each inseam; the notches on the inseam meet.



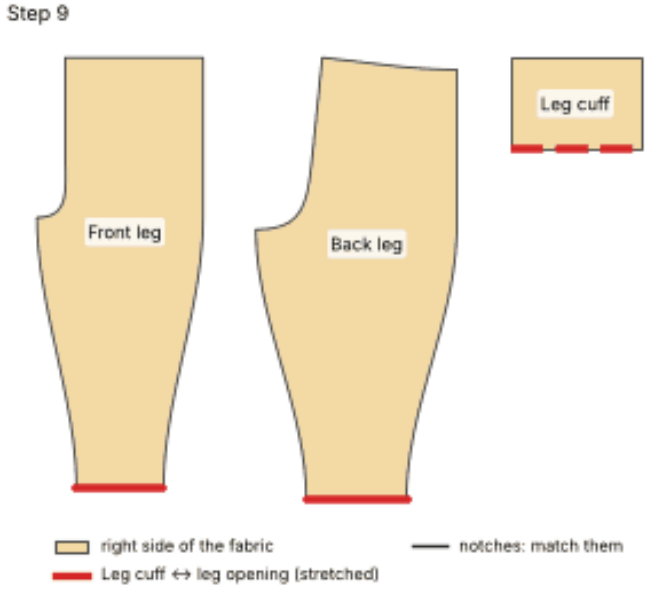
4. **Crotch seam:** turn one leg right side out and slip it into the other (wrong side out) so the legs lie right sides together. Sew the crotch seam in one go from the front to the back top edge; the inseams meet exactly.



5. **Waistband:** fold the waistband strip right sides together into a ring and sew the cross seam (centre back), leaving 3 cm open for the elastic. Fold lengthwise wrong sides together along the fold line.
6. Divide waistband and pants top edge into quarters and mark them (centre front/back, sides). Pin the waistband right sides together to the top edge (opening at the back, inside) and sew.



7. Thread the elastic, overlap the ends by 2 cm and sew them together, close the opening. Spread the gathers evenly and secure the elastic in the side seams by stitching in the ditch.
8. **Leg cuffs:** sew each cuff into a ring right sides together, then fold it lengthwise wrong sides together. Divide cuff and leg opening into quarters and mark them.
9. Slip the cuff over the right-side-out leg, put its seam at the inseam and sew it on while stretching. Press the seam allowance towards the pants.



10. Done! Press all seams (with sweat: steam, do not press down hard).

Validation report

Sewability check

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	66.0 cm	66.0 cm	+0.0 mm	□ OK	
Seam pairs	Inseam front ↔ back	41.7 cm	41.7 cm	+0.0 mm	□ OK	
Seam pairs	Front crotch seam (left ↔ right piece)	26.9 cm	26.9 cm	+0.0 mm	□ OK	
Seam pairs	Back crotch seam (left ↔ right piece)	31.3 cm	31.3 cm	+0.0 mm	□ OK	
Seam pairs	Waistband ↔ pants top edge	83.0 cm	83.0 cm	+0.0 mm	□ OK	

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Leg cuff ↔ leg opening (stretched)	70–85 % of 28.5 cm	20.0 cm = 70 %	-85.5 mm	✓ OK	rib band is stretched while sewing
Angles/transitions	Centre front ↔ waistband	90°	90.0°	+0.0°	✓ OK	
Angles/transitions	Centre back ↔ waistband	90°	90.0°	-0.0°	✓ OK	
Angles/transitions	Crotch ↔ inseam (front + back)	180°	180.9°	+0.9°	✓ OK	
Angles/transitions	Inseam ↔ hem front	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Hem ↔ side seam front	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Inseam ↔ hem back	90°	90.2°	+0.2°	✓ OK	
Angles/transitions	Hem ↔ side seam back	90°	90.2°	+0.2°	✓ OK	
Angles/transitions	Side seam ↔ waistband back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Side seam ↔ waistband (front + back)	180°	180.1°	+0.1°	✓ OK	
Measurements	Hip circumference pattern = body + ease	85.8 cm	85.8 cm	+0.0 %	✓ OK	body 66.0 cm + +30 % ease
Measurements	Top edge fits over the hip	≥ 67.3 cm	95.4 cm	+281.1 mm	✓ OK	top edge incl. usable stretch ≥ hip
Measurements	Front rise plausible	14.6 cm – 31.4 cm	27.3 cm	+0.0 mm	✓ OK	body rise 18.0 cm, harem drop 7.3 cm
Measurements	Inseam length plausible	39.0 cm – 60.5 cm	55.5 cm	+0.0 mm	✓ OK	body inseam 54.0 cm
Measurements	Foot fits through the leg opening	≥ 24.1 cm	29.5 cm	+54.8 mm	✓ OK	heel-instep circumference ≈ 1.35 × ankle
Geometry	Front leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Front leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Back leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Back leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Waistband (elastic): outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Waistband (elastic): grainline & cutting	present	cut 1 on fold		✓ OK	Main fabric
Geometry	Leg cuff: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Leg cuff: grainline & cutting	present	cut 2		✓ OK	Rib knit
Geometry	Waistband (elastic): cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		✓ OK	Centre front
Labels	Front leg: notches	on the cutting line	3 notch mark(s)		✓ OK	

Area	Check	Target	Actual	Δ	Result	Note
Labels	Back leg: notches	on the cutting line	6 notch mark(s)		□ OK	

Result: 31 OK, 0 warning(s), 0 error(s): the pattern is sewable.

Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 858}}`).

Name	Value	Derived from	Note
`hip_width`	85.8 cm	hip × (1 + 30 % ease)	
`front_back_offset`	0.5 cm	front quarter narrower, back quarter wider by this amount	
`front_width`	20.9 cm	hip_width / 4 − front_back_offset	
`back_width`	21.9 cm	hip_width / 4 + front_back_offset	
`front_crotch_extension`	4.3 cm	5 % of hip_width	
`back_crotch_extension`	8.6 cm	10 % of hip_width	
`rise`	24.3 cm	(side length − inseam) + seat ease + harem drop − waistband height	
`inseam_finished`	48.2 cm	inseam − allowance to the floor − harem drop + blousing	
`hem_width`	28.5 cm	cuff length / cuff stretch ratio	

Cutting layout · Harem pants size 122

